AYUSH BHARDWAJ

AI/ML Engineer | Generative AI · LLM Agents · RAG
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PROFILE SUMMARY

AI/ML undergraduate passionate about building intelligent, production-oriented AI systems across Generative AI, LLMs, RAG, AI Agents, and MCP. Hands-on experience developing LangGraph-based agents, MCP integrations, SQL agents, n8n workflow automation, and LLM applications, with a strong foundation in Python, machine learning, deep learning, NLP, and backend development. Driven by curiosity and a builder mindset, with a focus on turning emerging AI technologies into practical, reliable solutions.

EDUCATION

University School of Automation and Robotics, GGSIPU 2024 – 2028
B.Tech, Computer Science – AI-ML Specialization | GPA: 8.89

SKILLS

Languages: Python, C++, SQL, DSA
ML / Deep Learning: Supervised Learning, Neural Networks, CNNs, LSTMs, Transfer Learning, Generative AI
Agentic AI / LLMs: LangGraph, LangChain, RAG Pipelines, Tool Calling, MCP (Model Context Protocol), Groq, Ollama, n8n, AI Workflow Automation, Prompt Engineering
MLOps & Deployment: MLflow, DagsHub, DVC, Docker, CI/CD (GitHub Actions), AWS (S3, EC2, ECR), FastAPI
Data & Vector Stores: FAISS, ChromaDB, Vector Embeddings, SQL, SQLite, MongoDB
ML Workflow: EDA, Preprocessing Pipelines, Feature Scaling & Encoding, Data Visualisation, Model Evaluation
Tools & Platforms: Git, VS Code, Jupyter Notebook, Google Colab, Kaggle, Streamlit, n8n

PROJECTS

Network Security Threat Detection — ML Pipeline with MLOps *(MLflow · DagsHub · Docker · AWS (S3/ECR/EC2) · FastAPI · MongoDB)* — GitHub

- Built a full production-style ML pipeline (ingestion → validation → transformation → training → serving) for phishing/network-threat classification, with schema-drift checks on live data pulled from MongoDB Atlas
- Tracked experiments and registered models with MLflow on DagsHub for full run comparison and reproducibility
- Set up CI/CD with GitHub Actions: automated lint/test on push, Docker image build, and deployment to AWS ECR, served via a self-hosted EC2 runner pulling model artifacts from S3 at runtime
- Exposed predictions through a FastAPI service, containerized with Docker for consistent, repeatable deployment

SQL Agent — Ask Your Data *(LangGraph · DuckDB · FAISS · LangSmith · Streamlit)* — GitHub

- Built an agentic text-to-SQL system where users upload any CSV and query it in plain English, with a LangGraph pipeline engineered around the three real failure modes of text-to-SQL — schema hallucination, unsafe/incorrect SQL, and ambiguous questions — rather than a thin LLM wrapper
- Implemented column profiling with FAISS-based retrieval to ground SQL generation in relevant schema context, a keyword-based intent guard and SQL validator (SELECT-only, unknown table/column rejection) to block destructive or hallucinated queries, and a self-correction retry loop that regenerates SQL from real database errors
- Designed a human-in-the-loop ambiguity-check node using LangGraph's interrupt() and a checkpointer to pause and resume execution on genuinely ambiguous questions instead of guessing an interpretation
- Instrumented full observability with LangSmith (per-node traces, retry attempts, interrupt/resume cycle, latency and token usage); deployed live on Streamlit Community Cloud

YouTube RAG Chatbot *(RAG · LangChain · Ollama · ChromaDB)* — GitHub

- Built an end-to-end RAG system enabling users to chat with any YouTube video using local LLMs with no external API keys
- Implemented transcript ingestion with youtube-transcript-api, chunking, ChromaDB vector storage, and LangChain retrieval
- Integrated Ollama Llama 3.1 8B and nomic-embed-text for fast, context-aware Q&A over video content

N8N Personal AI Assistant *(n8n · AI Agent · Groq · Tavily · Google Workspace APIs)* — GitHub

- Built an action-driven personal assistant in n8n that turns natural-language requests into real actions across Gmail, Google Calendar, Tasks, Docs (notes), and Sheets (expense tracking), triggered through a webhook endpoint
- Designed an AI Agent with Groq LLM reasoning, conversational memory, and Tavily web search that selects the right tool per intent, with a calculator tool for exact expense totals
- Added agent safety rules: no fabricated tool results, append-only notes, and deletions only on explicit request

Chatbot-LangGraph *(LangGraph · Streamlit · Groq · FAISS · SQLite)* — GitHub

- Built a production-style conversational AI chatbot with LangGraph StateGraph, Groq Llama 3.3 70B, Streamlit UI, and SQLite checkpointing for persistent threaded chat history
- Implemented PDF-based RAG using PyPDFLoader, RecursiveCharacterTextSplitter, Ollama nomic-embed-text embeddings, and per-thread FAISS retrievers to keep document context isolated by session
- Added agentic tools for web search, stock lookup, calculator operations, token streaming, and multi-thread session management through LangGraph's tool routing loop